

Name: _____

Math 142 (Tully-Doyle)

Exam 2

September 21, 2026

Question	Points	Score
1	12	
2	30	
3	8	
Total:	50	

1. (12 points) Evaluate the limits:

(a) $\lim_{x \rightarrow 0} \frac{\arctan x}{x}$

(b) $\lim_{x \rightarrow \infty} x^{1/x}.$

2. (30 points) Evaluate the integrals:

(a)

$$\int \frac{x^2}{\sqrt{4-x^2}} dx$$

(b)

$$\int_1^4 \ln(\sqrt{x}) dx$$

(c)

$$\int 9 \tan^3 x \sec x \, dx$$

(d)

$$\int \frac{x}{x^2(x+1)} \, dx$$

3. (8 points) Determine if the integral converges or diverges. If it converges, find the limit.

$$\int_2^{\infty} \frac{\ln x}{x^2} dx$$

